

Goal: Build CRUD with PHP and MySQL

IMPORTANT

Our Goal Is CRUD

CRUD describes the four basic operations on stored data.

CRUD	SQL	What it does
Create	INSERT	Add a student
Read	SELECT	Find students
Update	UPDATE	Change a student
Delete	DELETE	Remove a student

PHP handles application logic. MySQL keeps the data.
PDO connects them.

One Application, Two Servers

Last week:

Browser → PHP server

This week:

Browser → PHP server → MySQL server
↓
persistent data

- The PHP server runs our application code.
- The MySQL server stores and retrieves data for many requests.

How We Reach CRUD

1. Run MySQL
- ↓
2. Create the database, table, and PHP account
- ↓
3. Connect PHP to MySQL with PDO
- ↓
4. Send SQL from PHP to create, read, update, and delete data

This preparation lesson completes step 1. The next two documents complete steps 2, 3, and 4.

IMPORTANT

Why MySQL Must Stay Running

MySQL is a separate server, not a library inside a PHP page.

- It stores data after a PHP request ends.
- It handles database files and SQL queries.
- PHP sends requests to MySQL and receives the results.

If MySQL is stopped, PHP can still run, but database operations cannot.

This is why MySQL runs as a **service** — a background program the operating system keeps alive so it's always there when PHP needs it.

- The concept and the exact `start` / `status` / `stop` / `restart` commands for your OS are in `course_tools/src/04. mysql installation.md`
- re-read the "MySQL as a Service" slides there if you need a refresher.

Before You Continue: Confirm MySQL Is Ready

This document assumes MySQL is **already installed and running** —

that was HW1 / `course_tools/src/04. mysql installation.md`.

Quick check:

```
mysql --version
```

Ubuntu / WSL2:

```
sudo service mysql status
```

macOS:

```
brew services list
```

If either command shows MySQL is missing or not running, stop here and go back to [course_tools/pdf/04. mysql installation.pdf](#) (or [course_tools/code/04. mysql installation/run.sh](#)) and follow the install/service steps again before continuing.

Open the MySQL Shell

In your terminal:

WSL2/Ubuntu:

```
sudo mysql
```

macOS:

```
mysql -u root
```

Inside the MySQL shell, the prompt appears as

`mysql>`. To exit back to your terminal, type `EXIT;`.

For password and shell-usage details, refer to

`course_tools/pdf/04. mysql`

`installation.pdf`.

Installation Checkpoint

Before continuing, confirm all three:

1. MySQL is installed.
2. The MySQL service is running.
3. You can open and exit the MySQL shell.

Next, we will create the persistent data used by our CRUD application.